WORKED SOLUTION

STEP II 2011, Question 7

Pure • Sequences and Series

STEP II 2011, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q7

The key idea

Never work with the sequences term by term: split every sum into a sum of powers of λ and a sum of powers of μ, and use the geometric series formula on each. The two numbers satisfy $\lambda+\mu=2$ and $\lambda\mu=-1$, so they are the roots of $x^2=2x+1$. That single quadratic makes all the awkward denominators collapse. The other tool is squaring: b_k^2 and a_k^2 both turn into a_{2k} plus a constant.

Solution

First, some facts we will use throughout:

$$\lambda-1=\sqrt{2},\qquad \mu-1=-\sqrt{2},\qquad \lambda\mu=1-2=-1,\qquad \lambda-\mu=2\sqrt{2}.$$

Also $a_0=2$ and $b_0=0$.

Part (i)

Since $\lambda\neq 1$ and $\mu\neq 1$, we can use the geometric series formula on each sum:

$$\sum_{r=0}^{n}\lambda^r=\frac{\lambda^{n+1}-1}{\lambda-1}=\frac{\lambda^{n+1}-1}{\sqrt{2}},\qquad \sum_{r=0}^{n}\mu^r=\frac{\mu^{n+1}-1}{\mu-1}=-\frac{\mu^{n+1}-1}{\sqrt{2}}.$$

Subtracting the second sum from the first,

$$\sum_{r=0}^{n}b_r=\frac{\lambda^{n+1}-1}{\sqrt{2}}+\frac{\mu^{n+1}-1}{\sqrt{2}}=\frac{a_{n+1}-2}{\sqrt{2}}=-\sqrt{2}+\frac{1}{\sqrt{2}}\,a_{n+1},$$

as required. Adding the two sums instead, the -1's cancel:

$$\boxed{\sum_{r=0}^{n}a_r=\frac{\lambda^{n+1}-1}{\sqrt{2}}-\frac{\mu^{n+1}-1}{\sqrt{2}}=\frac{1}{\sqrt{2}}\,b_{n+1}.}$$

(Quick test: with $n=0$ this says $a_0=b_1/\sqrt{2}=2\sqrt{2}/\sqrt{2}=2$, which is correct.)

Part (ii)

By part (i), the inner sum is $b_{m+1}/\sqrt{2}$. So

$$\sum_{m=0}^{2n}\left(\sum_{r=0}^{m} a_r\right) = \frac{1}{\sqrt{2}}\sum_{m=0}^{2n} b_{m+1} = \frac{1}{\sqrt{2}}\sum_{s=1}^{2n+1} b_s = \frac{1}{\sqrt{2}}\sum_{s=0}^{2n+1} b_s,$$

where the last step is allowed because $b_0 = 0$. Now part (i), with $2n+1$ in place of n, gives

$$\sum_{m=0}^{2n}\left(\sum_{r=0}^{m} a_r\right) = \frac{1}{\sqrt{2}}\left(-\sqrt{2} + \frac{a_{2n+2}}{\sqrt{2}}\right) = \tfrac{1}{2}a_{2n+2} - 1. \qquad (*)$$

Ask yourself How do we connect a_{2n+2} to b_{n+1}^2?

We square, and use $\lambda\mu = -1$ for the cross term:

$$b_{n+1}^2 = \lambda^{2n+2} + \mu^{2n+2} - 2(\lambda\mu)^{n+1} = a_{2n+2} - 2(-1)^{n+1}, \qquad a_{n+1}^2 = a_{2n+2} + 2(-1)^{n+1}.$$

If n is odd, then $n+1$ is even, so $b_{n+1}^2 = a_{2n+2} - 2$. Then $(*)$ equals $\frac{1}{2}(a_{2n+2} - 2) = \frac{1}{2}b_{n+1}^2$, as required.

If n is even, then $(-1)^{n+1} = -1$, so it is now a_{n+1}^2 that equals $a_{2n+2} - 2$. Hence

$$\boxed{n \text{ even}: \quad \sum_{m=0}^{2n}\left(\sum_{r=0}^{m} a_r\right) = \tfrac{1}{2}a_{n+1}^2}$$

or, equivalently, $\frac{1}{2}b_{n+1}^2 - 2$.

Part (iii)

By part (i), and then the squaring identity from part (ii),

$$\left(\sum_{r=0}^{n} a_r\right)^2 = \tfrac{1}{2}b_{n+1}^2 = \tfrac{1}{2}a_{2n+2} - (-1)^{n+1}.$$

For the second sum we again split into two geometric series, this time with ratios λ^2 and μ^2:

$$\sum_{r=0}^{n}\lambda^{2r+1} = \lambda\,\frac{\lambda^{2n+2} - 1}{\lambda^2 - 1}.$$

Ask yourself Can $\lambda^2 - 1$ be simplified?

Yes. Since $\lambda^2 = 2\lambda + 1$, we have $\lambda^2 - 1 = 2\lambda$, so the factor of λ cancels and the sum is $\frac{1}{2}(\lambda^{2n+2} - 1)$. Because μ satisfies the same quadratic, $\mu^2 - 1 = 2\mu$ and the μ-sum is $\frac{1}{2}(\mu^{2n+2} - 1)$ (note $\mu^2 \neq 1$, so the formula is valid). Adding the two,

$$\sum_{r=0}^{n} a_{2r+1} = \tfrac{1}{2}\left(a_{2n+2} - 2\right) = \tfrac{1}{2}a_{2n+2} - 1.$$

Therefore

$$\left(\sum_{r=0}^{n} a_r\right)^2 - \sum_{r=0}^{n} a_{2r+1} = \left(\tfrac{1}{2}a_{2n+2} - (-1)^{n+1}\right) - \left(\tfrac{1}{2}a_{2n+2} - 1\right) = 1 + (-1)^n.$$

If n is even this equals 2, as required. If n is odd it is 0, so

$$\boxed{n \text{ odd}: \quad \left(\sum_{r=0}^{n} a_r\right)^2 = \sum_{r=0}^{n} a_{2r+1}.}$$

Checks

The a_n are integers satisfying $a_{n+1} = 2a_n + a_{n-1}$: $2, 2, 6, 14, 34, 82, 198, 478, \ldots$. For part (ii) with $n = 3$, the partial sums $2, 4, 10, 24, 58, 140, 338$ add up to 576. Since $b_4 = 24\sqrt{2}$, we get $\frac{1}{2}b_4^2 = 576$, which matches. For part (iii) with $n = 2$: $(2 + 2 + 6)^2 - (2 + 14 + 82) = 100 - 98 = 2$. With $n = 1$: $(2 + 2)^2 - (2 + 14) = 0$. A computer checked all three parts, including both parities, for every n up to 24.

Common slips

- Forgetting that $\mu - 1 = -\sqrt{2}$ is negative, which swaps the roles of a and b in part (i).
- Writing $(\lambda\mu)^{n+1} = 1$ instead of $(-1)^{n+1}$; the sign is the whole point of the odd/even split.
- In part (ii), shifting the index of the sum of b_{m+1} without noticing that $b_0 = 0$ is what lets you start the sum at 0.

Beyond STEP

Both sequences satisfy the linear recurrence $x_{n+1} = 2x_n + x_{n-1}$, whose characteristic equation is $x^2 = 2x + 1$. So λ and μ are its roots, and a_n and b_n are particular solutions. Since $\lambda^n = \frac{1}{2}(a_n + b_n)$, the identity $\lambda^n\mu^n = (-1)^n$ becomes $a_n^2 - b_n^2 = 4(-1)^n$, a Pell-type equation in the ring $\mathbb{Z}[\sqrt{2}]$. Davenport's *The Higher Arithmetic* explains how such units and continued fractions for $\sqrt{2}$ fit together.

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